Minxing Zhang

Fourth-Year Ph.D. Candidate in Computer Science | Duke University
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Research focus. I make LLMs and AI agents trustworthy for real-world deployment: reliable, low-cost evaluation, privacy, and generalization. Now pushing into emerging frontiers: **agentic commerce** (AI agents that shop for users), **person distillation** (capturing a real person's expertise in AI), and LLMs that know when to ask questions.

Education

Duke University 2023–Present
Ph.D. in Computer Science *(Expected Graduation: December 2027)*
Durham, NC | Advisor: Professor Jian Pei

Emory University 2019–2022
B.S. in Computer Science and Applied Mathematics & Statistics
Atlanta, GA | GPA: 3.984/4.00; Highest Honor | Advisors: Professor Li Xiong and Professor Liang Zhao

Work and Internship Experience

Amazon Web Services | Applied Scientist Intern — Summer 2026

Amazon Bedrock Guardrails, Deep Engine-Science | Seattle, WA — *Research paper in submission*

Developed **Judge2Vec**, which turns each LLM judge into a computable embedding of its competence across evaluation dimensions, without human labels, to assemble small, complementary juries that are both reliable and cheap.

Showed that less is more: a jury of just 4 of 19 LLMs **beat querying all 19 at 83.9% lower cost**, achieving **state-of-the-art** results on 8 text and agentic benchmarks.

Also handles brand-new tasks with zero task data (98.5% of performance), any cost budget, and multimodal inputs.

AT&T Chief Data Office | Graduate Research Intern — Summer 2025

Bedminster, NJ | Mentors: Divesh Srivastava and Yaron Kanza — *Research paper in submission*

Enabled LLMs to **forget a user's private data on request without retraining** (GDPR/CCPA): the model learns only public data, private data sits in a separate deletable database (kNN-LM), and an adaptive selector picks which source answers.

Showed that deleting the data is not enough: the retrieval embeddings decide both answer quality and whether deleted facts leak. Designed embeddings that stay accurate yet **reveal nothing once deleted**.

Scaled to **19.6M entries (300 GB)** with user-level deletion; 1,000× more data made answers only 4.9× slower.

Microsoft & Emory University | DataThink Research Project — Summer 2022

Atlanta, GA

Built XGBoost rental-price prediction and SKATER spatial clustering models to analyze metro Atlanta's rental market.

Selected Research Projects

Agentic commerce. As AI agents begin to search and buy on users' behalf, marketplaces see only the agents' actions, never the users' intent. Formulated how platforms should serve these agents; building an LLM-based policy that predicts agent behavior and optimizes revenue across multi-turn interactions.

Person distillation. Laid the foundations for distilling a real person into AI: turning their documents, messages, and decisions into a model of their expertise and judgment, not just their style, so AI agents and "digital coworkers" can carry their know-how forward. Wrote **the field's first comprehensive framework**: a four-stage lifecycle (evidence, representations, methods, evaluation), six method families, and tests of whether AI captures the person, not the role.

LLM-based agent generalizability. Tackled a key question for deploying agents: will an agent that works in one setting still work in the next? Defined generalizability across instructions, tasks, environments, and domains, and mapped how to evaluate and improve it; published in **ACM Computing Surveys** and grew into two tutorials at KDD and PAKDD 2026.

MPCEval. Built a benchmark for AI-generated group conversations, such as meetings and team chats, that scores who speaks next, what they say, and whether the two fit, with no reference answers needed; evaluated diverse generation methods on collaborative problem-solving, scripted dialogue, and real enterprise discussions. **KDD 2026 oral.**

Privacy attacks and defenses. Showed how private data can leak through knowledge distillation (membership inference on teacher–student models) and across sequential data releases (a learned attack using hidden Markov models and reinforcement learning), and proposed mitigations.

Technical Skills

LLMs & agents: LLM evaluation (LLM-as-a-judge), benchmark design, post-training, RAG, AI agents
Trustworthy AI: differential privacy, unlearning | **Tools:** Python, PyTorch, Hugging Face, FAISS, AWS (Bedrock), Bash

Publications

Published and Accepted * Equal contribution

[1] MPCEval: A Benchmark for Multi-Party Conversation Generation
Minxing Zhang*, Yi Yang*, Zhuofan Jia, Xuan Yang, Jian Pei, Yuchen Zang, Xingwang Deng, Xianglong Chen
ACM SIGKDD, Datasets and Benchmarks Track; **oral presentation**. [Paper] [Code] 2026

[2] Generalizability of Large Language Model-Based Agents: A Comprehensive Survey
Minxing Zhang, Yi Yang, Roy Xie, Bhuwan Dhingra, Shuyan Zhou, Jian Pei
ACM Computing Surveys, 58(10), Article 263, 44 pages. [Paper] 2026

[3] On Membership Inference Attacks in Knowledge Distillation
Ziyao Cui*, **Minxing Zhang***, Jian Pei
ACM SIGKDD Explorations Newsletter, 28(1), pp. 32–40. [Paper] 2026

[4] Trustworthy LLM-Based Agents for Data-Centric AI
Tianlong Chen, Jian Pei, **Minxing Zhang**, Liang Zhao
ACM SIGKDD, Tutorial, Vol. 2, pp. 13299–13303; authors in alphabetical order. [Website] 2026

[5] Protecting Data Buyer Privacy in Data Markets
Minxing Zhang, Jian Pei
IEEE Internet Computing, 28(4), pp. 14–20. [Paper] 2024

[6] CSGAN: Modality-Aware Trajectory Generation via Clustering-Based Sequence GAN
Minxing Zhang, Haowen Lin, Shun Takagi, Yang Cao, Cyrus Shahabi, Li Xiong
IEEE MDM, pp. 148–157. [Paper] 2023

[7] Deep Spatial Prediction via Heterogeneous Multi-Source Self-Supervision
Minxing Zhang, Dazhou Yu, Yun Li, Liang Zhao
ACM Transactions on Spatial Algorithms and Systems, 9(3). [Paper] 2023

[8] Deep Geometric Neural Network for Spatial Interpolation
Minxing Zhang, Dazhou Yu, Yun Li, Liang Zhao
ACM SIGSPATIAL; Best Poster Runner-Up. [Paper] 2022

Preprints

[9] Person Distillation: Foundations and Principles for Personalized AI
Minxing Zhang*, Yi Yang*, Jian Pei
Preprints.org, 202608.1530.v1. [Paper] [Website] [GitHub] 2026

[10] Learning to Attack: Uncovering Privacy Risks in Sequential Data Releases
Ziyao Cui*, **Minxing Zhang***, Jian Pei
arXiv:2510.24807. [Paper] 2025

In Submission

[11] Judge2Vec: Selecting Compact LLM Juries from Unlabeled Judgments 2026
Minxing Zhang, Soumya Sanyal, Haiyang Yu, Han Xie, Eti Rastogi, Yueyan Chen, Xiong Zhou, Shuai Wang, Jian Pei, Zheng Qi, Jonathan Wu

[12] Scalable Private Information Management in Large Language Models for Question Answering 2026
Minxing Zhang, Zhuofan Jia, Aritra Guha, Yaron Kanza, Jian Pei, Divesh Srivastava

Tutorials and Invited Talks

KDD | Tutorial Co-organizer: Trustworthy LLM-Based Agents for Data-Centric AI. [Website] 2026

PAKDD | Tutorial Co-organizer: Generalizability of Large Language Model-Based Agents. [Program] 2026

Samsung ML Reading Group | Invited Speaker 2024

Honors and Awards

Academic Excellence Award, Emory Computer Science 2023
Phi Beta Kappa, Gamma Chapter of Georgia 2023
Best Poster Runner-Up, ACM SIGSPATIAL 2022
Travel Grant, ACM SIGSPATIAL 2022